

# Project Cheap Adapter: one world, different coordinates

A bounded test of the Platonic Representation Hypothesis, converted into deployable linear adapters and a shared multi-encoder hub.

Project Documents

34 verdicts | 21 confirmed | 7 falsified | 4 inconclusive | 2 not meaningful

EXECUTIVE FINDING

The project confirms PRH in a narrow form: trained independent models share mainly local structure that linear maps make useful. It also bounds the claim: convergence is not global or universal, and visible geometry need not transfer.

RESEARCH PROGRAM

8 encoders | 2 shipped systems

1

Experiment A

GPT-2 vs Pythia-160M

CKA + layer-wise ridge on 10,000 passages tests whether independent language models share linearly accessible structure.

R<sup>2</sup> mean 0.737; peak 0.797

2

Experiment B

MobileCLIP to SigLIP 2

One 512x768 ridge matrix aligns phone and server image encoders and is exported as a 0.79 MB fp16 Core ML artifact.

R@5 = 95.3% of native ceiling

3

Cheap Adapter hub

Seven experts, one space

One entry map per encoder reaches a shared 512-d whitened hub, replacing quadratic pairwise translation with linear growth.

N maps, not N(N-1)

EVIDENCE LADDER - STRONGER CLAIMS REQUIRE STRONGER TESTS

|   |                   |                   |                   |
|---|-------------------|-------------------|-------------------|
| 1 | Shared neighbours | mutual k-NN       | strong locally    |
| 2 | Shared order      | distance-rank rho | 0.29 cross-modal  |
| 3 | Shared structure  | CKA / RSA         | partial           |
| 4 | Shared subspace   | ridge + retrieval | 88-97% of ceiling |
| 5 | Same geometry     | rotation only     | rejected: 0.038   |

The pattern is the result: models agree most on local neighbourhoods, less on global shape, yet become highly interoperable after a fitted linear stretch.

HEADLINE EVIDENCE

0.9183

MLP matches ridge in tested setup

0.592

vision adapter held-out R<sup>2</sup>

0.768

bge-SBERT raw shape rho

96.7%

ConvNeXt hub transfer / native

RELATION TO HUH ET AL.

Huh et al. linked alignment to scale and capability. Cheap Adapter tests the narrower mechanism: learned local structure is linearly usable across modalities and lineages, with held-out ConvNeXt at 96.7% of native. Limits: no universal or global identity; scaling is not monotonic beyond the tested range; the source-rank cliff is numerical.

# From one adapter to the Cheap Adapter hub

The same scientific idea becomes an architecture: freeze every encoder, fit one linear entry map per space, and reuse downstream components across models.

SYSTEM DESIGN

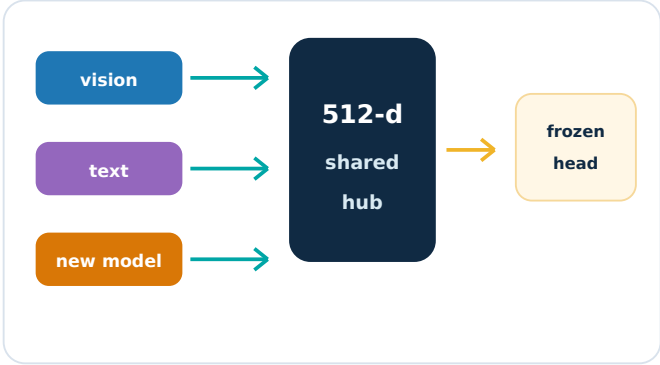
PAIRWISE

7 encoders -> 42 ordered maps

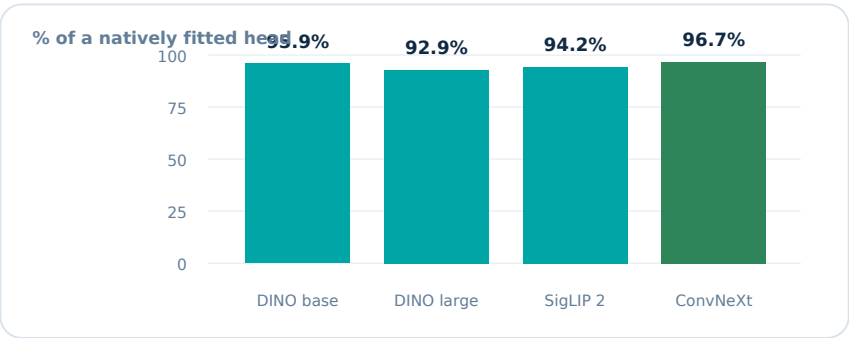
Adding one model requires maps to and from every existing model.

CHEAP ADAPTER HUB

7 encoders -> 7 entry maps



MEASURED TRANSFER - HELD-OUT 1,000-ITEM RETRIEVAL



WHAT TRANSFERRED

- A caption head trained on DINOv2-small serves encoders it never saw.
- SigLIP 2 tests independent lineage; ConvNeXt tests a different architecture.
- A new encoder still needs a fitted entry map: this is adaptation, not zero-shot.

THREE COMPLEMENTARY RESULTS

MAP-FREE

rho = 0.290

Mean cross-modal distance-rank agreement before any map; bge-SBERT reaches 0.768.

COMPOSED

103.6%

Mean hub performance versus bespoke pairwise maps across all 42 ordered pairs.

DEPLOYED

0.79 MB

The fp16 MobileCLIP-to-SigLIP matrix reaches 95.3% of native R@5.

WHY THE INFERENCE IS CREDIBLE

|                                                          |                                                                |                                                                   |                                                       |
|----------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------|
| <b>Frozen models</b><br>only closed-form maps are fitted | <b>Held-out scoring</b><br>8,533 train / 1,000 test in the hub | <b>Negative controls</b><br>shuffle and random maps sit at chance | <b>Nonlinear control</b><br>MLP ties linear at 0.9183 |
|----------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------|

PROJECT IN ONE PARAGRAPH

Two models do not share coordinates: direct cross-space retrieval is at chance without an adapter. Yet they encode related structure, and one phone-sized matrix makes it usable. What they share most strongly is local - nearby items agree more than the global shape. Coordinate frames are cheap to fix; semantic arrangement is what training buys. This does not imply one identical latent space.

# Fractal probe: the role of training data

FractalDB-only encoders provide a sharp control: structured training with zero natural photographs, bracketed by matched random-init and ImageNet twins.

## THREE TESTS, ONE CONSISTENT READING

G8

Visible geometry

Four map-free metrics reproduce nonzero cross-modal structure. Raw shape rho is 0.288 (0.329 without collapsed GPT-2); whitening lowers it to 0.062.

G9

Fractal vs controls

P1/P2 confirm real geometry above matched random controls; P5w confirms after whitening. The CNN signal survives PC1 removal; P3 asymmetry remains inconclusive.

G10

Functional recovery

H1 is falsified: fractal transfer stays at the floor and matches random twins. H3 CNN/ViT ratios are not meaningful under the floor guard.

## G9 - VISIBLE STRUCTURE RELATIVE TO MATCHED CONTROLS

Reality fraction R

0 = random architecture; 1 = ImageNet twin

Fractal CNN

0.09

Fractal ViT

0.21

Matched-norm CNN control: 0.067 fractal vs 0.036 random. The training effect survives preprocessing correction.

Robustness and uncertainty

- CNN image-vs-text R gap: 95% CI [-0.106, +0.100].
- ViT image-vs-text R gap: 95% CI [-0.053, +0.135].
- Corrected drop-1: CNN ordering survives; ViT falls below random.
- CNN-vs-ViT contrast is not interpreted because capacity differs.

## G10 - FROZEN HUB/HEAD RETRIEVAL (R@1; OPERATIONAL FLOOR = 0.050)

| ENCODER      | TRANSFER | NATIVE | RANDOM TWIN | INTERPRETATION                      |
|--------------|----------|--------|-------------|-------------------------------------|
| Fractal CNN  | 0.014    | 0.014  | 0.012       | at floor; no training-specific gain |
| Fractal ViT  | 0.020    | 0.022  | 0.017       | at floor; no training-specific gain |
| ImageNet CNN | 0.258    | 0.274  | -           | clear functional transfer           |
| ImageNet ViT | 0.204    | 0.208  | -           | clear functional transfer           |

## PROJECT VERDICT LEDGER - 34 PRE-REGISTERED PREDICTION UNITS

21

CONFIRMED

Includes learned geometry, cross-modal and cross-lineage transfer, and G9 P1/P2/P5w.

7

FALSIFIED

Includes H1 semantic recovery through the hub; it does not mean that all fractal effects failed.

4

INCONCLUSIVE

P3 image-vs-text reality asymmetry: two architectures across raw and whitened views.

2

NOT MEANINGFUL

H3 CNN and ViT percentage-of-native ratios are blocked by the 0.050 floor guard.

HONEST HEADLINE AND SCOPE

Confirmed narrowly: trained independent models share mainly local structure that linear maps make useful across modalities and lineages. Bounded: not global or universal; scale is not monotonic, natural-world data matters, and visible geometry need not transfer.